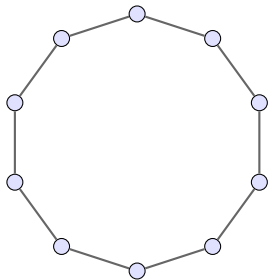
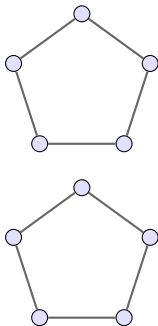


Decagon C_{10}



10 nodes, each of degree 2

Two pentagons $2 \times C_5$



10 nodes, each of degree 2

1-WL: in both graphs $\rightarrow$ every node gets the same color at *every* step
 $\Rightarrow$ **no 1-WL MPNN** can distinguish C_{10} from $2 \times C_5$